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Summary of problem

FuturePower Inc. is evaluating the possibility of building a 100 MW gas turbine plant fueled exclusively on green hydrogen. In order to determine the project's financial viability, the Levelized Cost of Electricity (LCOE) needs to be determined. This plant is expected to operate with an 80% capacity factor over a 20-year lifespan. This analysis includes fixed and variable operating and maintenance costs, hydrogen fuel costs, efficiency losses, and the time value of money using a 15% discount rate. A stochastic analysis using @RISK in Excel was performed to identify the variables most sensitive to cost variability.

Methods and Model

I used Excel to include a given data table with capital cost, O&M costs, hydrogen cost, efficiency, and other plant parameters. I created equations embedded to calculate hydrogen demand, cost breakdowns, and total costs, and incorporated the discount factor calculations and formulas to show cost over time. A Monte Carlo simulation using @RISK to help identify the most important variables that influence the LCOE.

Key information used:

- Capital Expenditure (CAPEX): 120 million
- Fixed Operating and Maintenance (O&M) Costs: $3.5 million per year
- Variable O&M Costs: $1.50 per MWh of electricity produced
- Fuel Cost (Hydrogen): $7.50 per kg
- Heating Value (HV) of Hydrogen: 130 MJ/kg
- Combined Plant Efficiency: 60%
- Plant Capacity Factor: 80%
- Project Lifetime: 20 years
- Discount Rate: 15%

Equations used:

- Annual Energy Generation (MWh) = 100 MW * 8760 hr/year * 0.80
- Hydrogen Required per Year (kg) = (Annual Energy * 3.6) / (130 MJ/kg * Efficiency)
- Total Annual Costs = Fixed O&M + Variable O&M + Fuel Costs
- LCOE ($/MWh) = Present Value of Total Costs / Present Value of Total Electricity

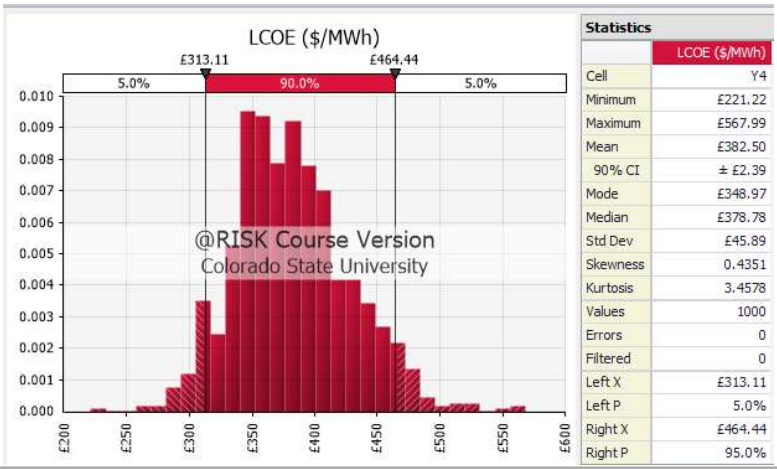
Given			Calculations															
Capital Expenditure (CAPEX)	$120,000,000			Annual Electricity Generation		Annual Hydrogen Consumption			Levelized Cost of Electricity									
Fixed Operating and Maintenance (O&M) Costs	$3,500,000	per year	Years	Electricity Generation [MWh/year]	PV_Electricity Generation [MWh]	Energy Output [MJ]	Efficiency [MJ]	Mass [kg]	CAPEX	Fixed O&M	PV_Fixed O&M	Variable O&M	PV_Variable O&M	Fuel Cost	PV_Fuel Cost	PV_Total	LCOE	
Variable O&M Costs	$1.50	per MWh	0						$120,000,000							$120,000,000.00	$380.00	
Fuel Cost (Hydrogen)	$7.50	per kg	1	700,800	609,391	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$3,043,478.26	$1,051,200	$914,086.96	$242,584,615.38	$210,943,143.81	$214,900,709.03		
Heating Value (HV) of Hydrogen	130	MJ/kg	2	700,800	529,905	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$2,646,502.84	$1,051,200	$794,858.22	$242,584,615.38	$183,428,820.71	$186,870,181.77		
Combined Plant Efficiency	60%		3	700,800	460,787	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$2,301,306.81	$1,051,200	$691,181.06	$242,584,615.38	$159,503,322.35	$162,495,810.23		
Plant Capacity Factor	80%		4	700,800	400,685	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$2,001,136.36	$1,051,200	$601,027.01	$242,584,615.38	$138,698,541.18	$141,300,704.55		
Project Lifetime	20	years	5	700,800	348,421	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$1,740,118.57	$1,051,200	$522,632.18	$242,584,615.38	$120,607,427.11	$122,870,177.87		
Discount Rate	15%		6	700,800	302,975	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$1,513,146.59	$1,051,200	$454,462.77	$242,584,615.38	$104,876,023.57	$106,843,632.93		
Gas turbine capacity	100	MW	7	700,800	263,457	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$1,315,779.64	$1,051,200	$395,185.02	$242,584,615.38	$91,196,542.24	$92,907,506.89		
			8	700,800	229,093	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$1,144,156.21	$1,051,200	$343,639.14	$242,584,615.38	$79,301,341.08	$80,789,136.43		
			9	700,800	199,211	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$994,918.44	$1,051,200	$298,816.65	$242,584,615.38	$68,957,687.89	$70,251,422.98		
			10	700,800	173,227	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$865,146.47	$1,051,200	$259,840.56	$242,584,615.38	$59,963,206.86	$61,088,193.90		
			11	700,800	150,632	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$752,301.28	$1,051,200	$225,948.32	$242,584,615.38	$52,141,919.01	$53,120,168.61		
			12	700,800	130,985	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$654,175.03	$1,051,200	$196,476.80	$242,584,615.38	$45,340,799.14	$46,191,450.96		
			13	700,800	113,900	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$568,847.85	$1,051,200	$170,849.39	$242,584,615.38	$39,426,781.86	$40,166,479.10		
			14	700,800	99,043	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$494,650.30	$1,051,200	$148,564.69	$242,584,615.38	$34,284,158.14	$34,927,373.13		
			15	700,800	86,124	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$430,130.70	$1,051,200	$129,186.68	$242,584,615.38	$29,812,311.43	$30,371,628.81		
			16	700,800	74,891	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$374,026.69	$1,051,200	$112,336.25	$242,584,615.38	$25,923,749.07	$26,410,112.01		
			17	700,800	65,122	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$325,240.60	$1,051,200	$97,683.69	$242,584,615.38	$22,542,390.49	$22,965,314.79		
			18	700,800	56,628	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$282,817.92	$1,051,200	$84,942.34	$242,584,615.38	$19,602,078.69	$19,969,838.95		
			19	700,800	49,242	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$245,928.62	$1,051,200	$73,862.91	$242,584,615.38	$17,045,285.82	$17,365,077.34		
			20	700,800	42,819	2,522,880,000	4,204,800,000	32,344,615.38		$3,500,000	$213,850.98	$1,051,200	$64,228.61	$242,584,615.38	$14,821,987.67	$15,100,067.26		

Figure 5: Excel spreadsheet used in calculations and @RISK data

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Questions

1. Calculate the annual electricity generation (in MWh) of the power plant and annual fuel consumption of hydrogen (in kg).
    a. The plant will produce the annual electricity generation of 700,800 MWh/year and the annual fuel consumption of hydrogen of 32,344,615.38 kg/year.
2. Calculate the levelized cost of electricity (LCOE) in $/MWh.
    a. Using an NPV based calculation, the LCOE was calculated to be $380.00/MWh.
3. Perform a stochastic analysis to identify the most important variables that significantly influence the LCOE of this hydrogen-fueled power plant.
    a. Using @RISK to create a Monte Carlo simulation, it showed the mean LCOE as $382.50/MWh, with the 90% confidence interval between $313.11 and $464.44, a standard deviation of $45.89, and a minimum and maximum of $221.22 and $567.99 respectively. The hydrogen cost was the most critical driver of LCOE variability. The @RISK analysis is shown in Figure 6 below.



Statistics	LCOE ($/MWh)
Cell	Y4
Minimum	£221.22
Maximum	£567.99
Mean	£382.50
90% CI	± £2.39
Mode	£348.97
Median	£378.78
Std Dev	£45.89
Skewness	0.4351
Kurtosis	3.4578
Values	1000
Errors	0
Filtered	0
Left X	£313.11
Left P	5.0%
Right X	£464.44
Right P	95.0%

Summary Statistics

Output	Cell	Graphs	Function	Minimum	Maximum	Mean	Std Dev	5%	95%
LCOE ($/MWh)	'Problem 4 @RISK (2)'!Y4		RiskOutput("LCOE ($/MWh)")	£221.22	£567.99	£382.50	£45.89	£313.11	£464.44

Figure 6: @RISK simulation results

Executive Summary

The LCOE of a 100 MW hydrogen-fueled gas turbine plant over a 20-year lifetime is estimated at $382.50/MWh on average, with high sensitivity to hydrogen fuel prices. Stochastic modeling shows potential LCOE ranging from $313.11 to $464.44/MWh.